AI AGENT CATALYST

Global Industry Hackathon 2026

Complete Hackathon Guide

All Milestones in One Notebook

MIT Session | Databricks Intelligence Platform

15 Teams | 15 Industries | 60 Minutes | 100 Points | 8 Milestones

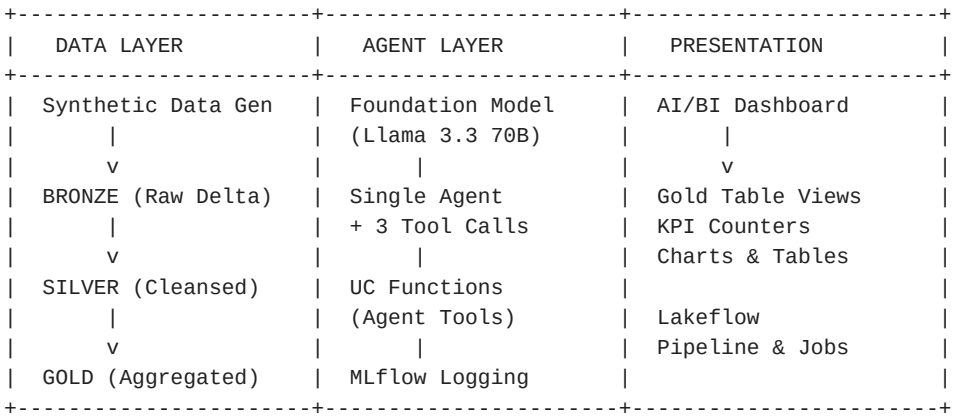
Scoring Overview

Milestone	Task	Points	Time	Notebook
Setup	Environment & team registration	-	5 min	02-Environment-Setup
M1: Bronze	Generate 3 Delta tables	10	7 min	03-Bronze-Data-Ingestion
M2: Silver	Cleanse & enrich data	10	7 min	04-Silver-Transformation
M3: Gold	KPI + analytics tables	10	7 min	05-Gold-Business-Metrics
M4: Agent	LLM + 3 tool-calling agent	20	12 min	06-AI-Agent-Development
M5: UCF	3 SQL UC Functions	10	6 min	07-Agent-UC-Functions
M6: Pipeline	Pipeline functions + Job	10	6 min	08-Pipeline-and-Jobs
M7: Dashboard	3 views + AI/BI dashboard	10	5 min	09-Dashboard-Creation
M8: E2E	End-to-end validation	10	3 min	10-End-to-End-Validation
Quiz	20 MCQs self-assessment	10	2 min	11-Self-Assessment-Quiz
TOTAL		**100 pts**	**60 min**	

Key Information

Catalog: `lakemeter_catalog`
Schema: `hackathon_{group_id}` (e.g. `hackathon_g01`)
LLM Model: `databricks-meta-llama-3-3-70b-instruct`
Auth: `No API key needed (serverless FMAPI)`
Scoring: `Auto-validated via 10-End-to-End-Validation`
Tiebreaker: `Agent quality + dashboard aesthetics + code cleanliness`

Architecture



Domain Assignments - 15 Teams

Group	Domain	Table 1 (Primary)	Table 2 (Events)	Reference
G01	Healthcare	bronze_patients	bronze_vitals	ref_departments
G02	Fintech & Banking	bronze_accounts	bronze_transactions	ref_branches
G03	E-commerce & Retail	bronze_orders	bronze_products	ref_categories
G04	Telecom	bronze_subscribers	bronze_tickets	ref_plans
G05	Manufacturing	bronze_production	bronze_inventory	ref_machines
G06	Insurance	bronze_policies	bronze_claims	ref_policy_types
G07	Pharma & Life Sci	bronze_trials	bronze_adverse_events	ref_compounds
G08	Automotive	bronze_vehicles	bronze_trips	ref_fleets
G09	Energy & Utilities	bronze_meters	bronze_outages	ref_tariffs
G10	Logistics	bronze_shipments	bronze_warehouses	ref_carriers
G11	Media	bronze_content	bronze_engagement	ref_genres
G12	Entertainment	bronze_events	bronze_bookings	ref_venues
G13	Travel & Hospitality	bronze_bookings	bronze_reviews	ref_hotels
G14	Education & EdTech	bronze_students	bronze_assessments	ref_courses
G15	Govt & Public	bronze_applications	bronze_budgets	ref_departments

SETUP - Environment Configuration

SETUP **- pts**

Time: 5 min

Objective

Configure team identity, create schema, register team.

Steps

1. Create group_id dropdown widget (G01-G15)
2. Build DOMAIN_MAP dictionary
3. Set GROUP_ID, DOMAIN, CATALOG="lakemeter_catalog", SCHEMA, FULL_SCHEMA
4. Run USE CATALOG / USE SCHEMA
5. CREATE SCHEMA IF NOT EXISTS
6. Create team_registration Delta table (1 row)
7. Print config

Key APIs

```
dbutils.widgets.dropdown(name, default, choices, label)
spark.sql(f"CREATE SCHEMA IF NOT EXISTS ...")
```

Checklist

- [ ] Widget group_id visible
- [ ] Variables printed correctly
- [ ] Schema exists
- [ ] team_registration has 1 row

M1: BRONZE **10 pts**

Time: 7 min

Objective

Generate domain-specific synthetic data and create 3 raw Delta tables.

Steps

1. Import random, datetime
2. Create helpers: gen_name(), gen_id(), rand_date()
3. Look up 3 table names from Domain Reference
4. Generate >=250 rows for Table 1
5. Generate >=250 rows for Table 2
6. Create Reference Table (5-10 rows)
7. Write each as Delta with saveAsTable()
8. Print row counts

Key APIs

```
spark.createDataFrame(list_of_tuples, column_names)
df.write.format("delta").mode("overwrite").saveAsTable()
```

Checklist

- [ ] Table 1: >=250 rows, Delta, correct columns
- [ ] Table 2: >=250 rows, Delta
- [ ] Ref Table: >=5 rows
- [ ] SELECT count(*) confirms counts

M2: SILVER — 10 pts

Time: 7 min

Objective

Cleanse and enrich bronze into analytics-ready silver tables.

Steps

1. Read bronze tables with spark.table()
2. Trim all string columns with F.trim()
3. Add: _ingested_at, _source_layer, _data_quality, _row_hash (MD5)
4. Add status_category: Active / Completed / Alert / In Progress
5. Write as silver_ prefixed Delta tables
6. Add COMMENT ON TABLE

Key APIs

```
from pyspark.sql import functions as F
F.when(...).when(...).otherwise(...)
```

Checklist

- [ ] Silver tables have all 5 metadata columns
- [ ] DISTINCT status_category shows 2-4 values
- [ ] Table comments set

M3: GOLD — 10 pts

Time: 7 min

Objective

Build aggregated KPI and analytics tables.

Steps

1. Read both silver tables
2. Compute >=5 KPIs: counts, averages, DQ rate
3. Build gold_kpi_summary with proper schema
4. Pick grouping column, build gold_analytics_detail
5. Write both as Delta + COMMENT ON TABLE

Key APIs

```
df.groupBy("col").count() | df.agg(F.avg("col"))
```

Checklist

- [ ] gold_kpi_summary >=5 rows
- [ ] gold_analytics_detail has data
- [ ] Both have COMMENT ON TABLE

M4: AI AGENT — 20 pts

Time: 12 min | HIGHEST VALUE

Objective

Build AI agent with tool-calling using Databricks Foundation Model API.

Steps

1. Initialize: client = mlflow.deployments.get_deploy_client("databricks")
2. Set MODEL = "databricks-meta-llama-3-3-70b-instruct"
3. Write SYSTEM_PROMPT with domain persona + tool descriptions
4. Define tools list (3 tools in OpenAI format)
5. Write tool_search_records -> queries silver, returns JSON
6. Write tool_get_kpi_metrics -> queries gold_kpi_summary
7. Write tool_get_analytics_detail -> queries gold_analytics_detail
8. Build run_agent(query): LLM -> tool -> execute -> LLM -> answer
9. Test with 2+ domain queries

Key APIs

```
client.predict(endpoint=MODEL, inputs={"messages":..., "tools":...})
df.toPandas().to_dict("records")
```

Checklist

- [ ] LLM client OK
- [ ] SYSTEM_PROMPT set
- [ ] 3 tool defs
- [ ] 3 tool functions return JSON
- [ ] run_agent() loop works
- [ ] Query 1: correct tool
- [ ] Query 2: different type

M5: UC FUNCTIONS — 10 pts

Time: 6 min

Objective

Register 3 agent tools as SQL functions in Unity Catalog.

Steps

1. CREATE FUNCTION fn_search_records(status_filter, max_results) RETURNS TABLE
2. CREATE FUNCTION fn_get_kpis(metric_name) RETURNS TABLE
3. CREATE FUNCTION fn_get_analytics(status, min_records) RETURNS TABLE
4. Add COMMENT to each
5. Verify with SELECT * FROM fn_xxx() LIMIT 5

Key APIs

```
CREATE OR REPLACE FUNCTION ... RETURNS TABLE RETURN SELECT ...
```

Checklist

- [ ] fn_search_records() returns data
- [ ] fn_get_kpis() returns data
- [ ] fn_get_analytics() returns data
- [ ] All have COMMENT

M6: PIPELINE & JOB — 10 pts

Time: 6 min

Objective

Create pipeline functions and a Lakeflow Job.

Steps

1. Write pipeline_bronze_to_silver()
2. Write pipeline_silver_to_gold()
3. Execute both and print results
4. Import WorkspaceClient from databricks.sdk
5. Create job: etl_pipeline -> agent_validation
6. Add tags: hackathon=2026, group=GROUP_ID

Key APIs

```
w.jobs.create(name=..., tasks=[...], tags={...})
```

Checklist

- [ ] Both pipeline functions run
- [ ] 3 pipeline KPIs added
- [ ] Job created (ID printed)
- [ ] Tags set

M7: DASHBOARD — 10 pts

Time: 5 min

Objective

Create 3 SQL views and build an AI/BI Dashboard.

Steps

1. CREATE VIEW vw_dashboard_kpis
2. CREATE VIEW vw_dashboard_analytics
3. CREATE VIEW vw_dashboard_detail
4. Create dashboard via SDK or UI

Key APIs

```
CREATE OR REPLACE VIEW ... AS SELECT ...
```

Checklist

- [ ] All 3 views return data
- [ ] Dashboard created

M8: VALIDATION — 10 pts

Time: 3 min

Objective

Run automated checks to validate all milestones.

Steps

1. Run 10-End-to-End-Validation notebook
2. Or run manual self-check in this notebook
3. Verify all tables, functions, agent, views exist

Checklist

- [ ] All bronze/silver/gold tables exist
- [ ] 3 UC functions registered
- [ ] Agent responds
- [ ] 3 dashboard views exist

What You Built

Layer	Deliverables
Bronze	2 primary + 1 reference = 3 Delta tables
Silver	2 enriched tables with metadata
Gold	KPI summary + analytics detail = 2 tables
UC Functions	3 SQL RETURNS TABLE functions
Views	3 dashboard SQL views
AI Agent	1 agent with LLM + 3 tools
Lakeflow Job	1 job with 2 tasks
Dashboard	1 executive AI/BI dashboard

Process Flow

```
SETUP -> BRONZE -> SILVER -> GOLD -> AGENT -> UCF -> PIPELINE -> DASHBOARD
(5 min)  (7 min)  (7 min)  (7 min) (12 min) (6 min)  (6 min)    (5 min)
```

Final Steps

> Run 10-End-to-End-Validation for automated scoring (10 pts)

> Complete 11-Self-Assessment-Quiz - 20 MCQs (10 pts)

> Raise your hand when done!

Tiebreaker: Agent quality + Dashboard aesthetics + Code cleanliness